Anqi Tan

Engineer · AI & Data for Health Technologies

anqitan2@illinois.edu | anqitan.github.io | LinkedIn | Google Scholar

Bioengineering Ph.D. candidate who builds data pipelines and LLM tools around real biomedical devices; two-time Ashby Prize hackathon winner (1st place, 2025).

EDUCATION

Ph.D. in Bioengineering, University of Illinois Urbana-Champaign Aug 2022 – Expected May 2027

Advisor: Prof. Brian T. Cunningham · Nanosensors Group

B.S. in Mechanical Engineering (dual degree), UIUC & Zhejiang University Aug 2018 – Jun 2022

EXPERIENCE

Graduate Research Assistant, Nanosensors Group, UIUC Sep 2022 – Present

- Developed the image-analysis pipeline for a pocket-size fluorescence reader: **automated test/control-line detection, background correction, and 4PL calibration** to convert strip images into cortisol concentrations in an Android app (Kotlin).
- Lead developer of the portable fluorescence reader for a **$4.87M DoD (PRMRP)** project on point-of-care sepsis/ARDS phenotyping with Stanford, WashU, and UCSF.
- Developed a plasmonic-fluor lateral flow assay for salivary cortisol (**≤0.12 ng/mL** LOD; **153 adults** sampled); **oral presentation, APHA 2026**.

Undergraduate Research Leader, ZJU-UIUC Institute 2019 – 2021

- Led a national-level research project designing a re-localizable remote-center-of-motion mechanism for minimally invasive surgery (**first-author paper, IEEE ICARM 2021**).

SELECTED PROJECTS

CRISPR Mechanism Explainer (Ashby Prize Hackathon, NCSA) — 1st Place Oct 2025

- Two-person team; found label bias in the public benchmark, built a balanced validation set with experimental-condition context, and fine-tuned an LLM (Qwen3) to classify CRISPR screen outcomes and explain the mechanism; shipped a Gradio chat demo.

LLM-Automated Gravitational-Wave Workflow (Ashby Prize Hackathon) — Honorable Mention Apr 2024

- Five-person team; built a streamlined pipeline using OpenAI services as a front-end to classify gravitational-wave signals from black-hole collisions.

PUBLICATIONS

First author: IEEE ICARM 2021 paper · APHA 2026 oral presentation · pocket fluorescence reader manuscript (in preparation). **Co-author:** 4 journal papers (*Chemical Reviews*, *Lab on a Chip*, *ACS Nano Medicine*, *Biosensors*) and 1 Elsevier book chapter.

HONORS & AWARDS

1st Place, Ashby Prize Computational Science Hackathon, NCSA (2025) · Honorable Mention, Ashby Prize Hackathon (2024) · National Silver Medal, China "Internet+" Innovation & Entrepreneurship Competition (2021)

TEACHING

TA, BIOE/ECE 415 Biomedical Instrumentation Lab (Fall 2025) · TA, ME 170 Computer-Aided Design (3 semesters)

SKILLS

AI & Data: Python, Android app development (Kotlin), LLM fine-tuning (Qwen3), OpenAI API, Gradio, image processing, curve fitting (4PL), MATLAB

Hardware: Raspberry Pi, ESP32, Arduino, LabVIEW, sensors (IMU), optics

Biosensing: lateral flow assays (competitive & sandwich), plasmonic-fluor labels, immunoassay optimization, ELISA, saliva/plasma sample prep, 3D cell culture, fluorescence imaging

Languages: English (fluent), Chinese (native)